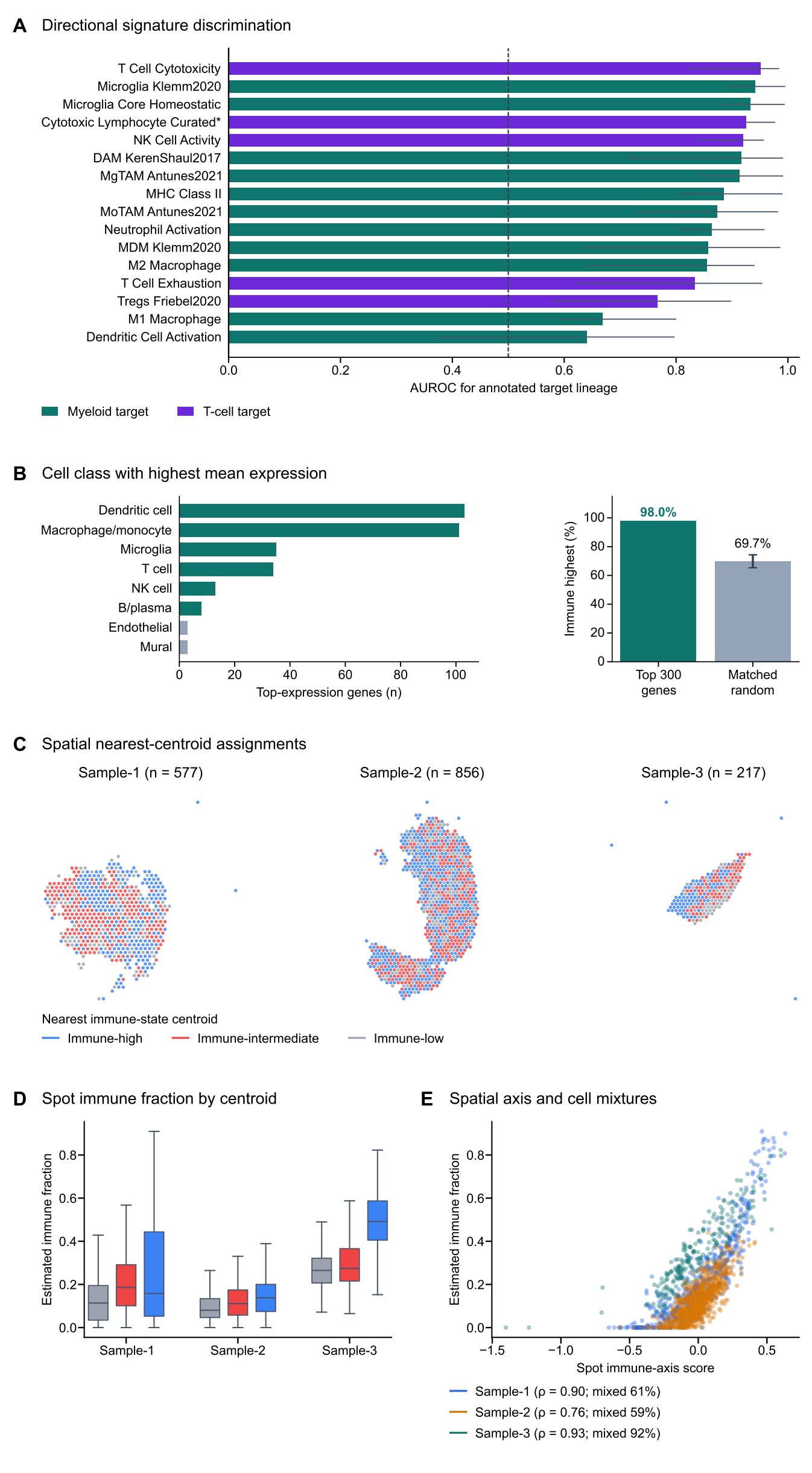

A
Directional signature discrimination
T Cell Cytotoxicity
Microglia Klemm2020
Microglia Core Homeostatic
Cytotoxic Lymphocyte Curated*
NK Cell Activity
DAM KerenShaul2017
MgTAM Antunes2021
MHC Class II
MoTAM Antunes2021
Neutrophil Activation
MDM Klemm2020
M2 Macrophage
T Cell Exhaustion
Tregs Friebel2020
M1 Macrophage
Dendritic Cell Activation
0.0
0.2
0.4
0.6
0.8
1.0
AUROC for annotated target lineage
Myeloid target
T-cell target
B
Cell class with highest mean expression
Dendritic cell
Macrophage/monocyte
Microglia
T cell
NK cell
B/plasma
Endothelial
Mural
0
20
40
60
80
100
Top-expression genes (n)
98.0%
69.7%
Immune highest (%)
Top 300 genes
Matched random
C
Spatial nearest-centroid assignments
Sample-1 (n = 577)
Sample-2 (n = 856)
Sample-3 (n = 217)
Nearest immune-state centroid
Immune-high
Immune-intermediate
Immune-low
D
Spot immune fraction by centroid
Estimated immune fraction
Sample-1
Sample-2
Sample-3
E
Spatial axis and cell mixtures
Estimated immune fraction
Spot immune-axis score
Sample-1 (ρ = 0.90; mixed 61%)
Sample-2 (ρ = 0.76; mixed 59%)
Sample-3 (ρ = 0.93; mixed 92%)